Supplementary Notes: Energy-Based Models

Hong Qin, Old Dominion University

Fall 2026

About these supplementary notes

Energy-based models (EBMs) were covered as a full lecture in the 16-week version of this course. In the 8-week format they are not part of the core Week 5 material and are provided here as optional reading. They are a natural companion to the diffusion lecture notes: the Langevin sampler described below is the ancestor of iterative denoising, and the score-matching view of diffusion (Week 5, Topic 2) is precisely a way of learning the gradient of an energy landscape without ever computing the partition function.

1 The Core Idea: Assigning Energy

An **energy function** $E_\theta(x)$ assigns a score to each possible configuration x of the data:

- Low energy → "good fit," likely.
- High energy → "bad fit," unlikely.

Analogy: Imagine a ball rolling on a hilly landscape:

- Valleys = low energy = ball tends to settle here.
- Peaks = high energy = ball avoids these areas.

2 From Energy to Probability

The connection between energy and probability is given by the Boltzmann distribution:

$$p_\theta(x) = \frac{\exp(-E_\theta(x))}{Z_\theta},$$

where $Z_\theta = \int \exp(-E_\theta(x))\, dx$ ensures probabilities sum to 1.

Analogy: Imagine a room with snacks of different attractiveness:

- People gather in corners with better snacks (lower energy).
- Z_θ ensures that all probabilities sum to 100%.

3 Training: Shaping Valleys

Training shapes the energy landscape so valleys align with real data:

- Lower the energy of real data (dig valleys).
- Raise the energy of fake data (build hills).

Analogy: Like a sculptor shaping clay:

- Real data $\rightarrow$ press down.
- Fake data $\rightarrow$ push up.

4 The Partition Function Problem

Computing Z_θ directly:

$$Z_\theta = \int \exp(-E_\theta(x))\, dx$$

is intractable for high-dimensional data. Hence, we use approximation and sampling.

5 Sampling: Walking Around the Landscape

We use stochastic sampling like Langevin dynamics:

$$x_{t+1} = x_t - \frac{\eta}{2}\nabla_x E_\theta(x_t) + \sqrt{\eta}\,\epsilon_t$$

with noise $\epsilon_t \sim \mathcal{N}(0, I)$.

Analogy: Blindfolded walking:

- Use the slope (gradient) to walk downhill.
- Add random stumbles to explore other valleys.

6 Positive vs Negative Phase

The gradient of the log-likelihood is:

$$\nabla_\theta \mathcal{L} = -\mathbb{E}_{x\sim p_{\text{data}}}[\nabla_\theta E_\theta(x)] + \mathbb{E}_{x\sim p_\theta}[\nabla_\theta E_\theta(x)]$$

- Positive phase: lower energy of real data.
- Negative phase: raise energy of fake samples.

Analogy: Like playing Whac-A-Mole:

- Real data = dig holes.
- Fake data = hammer them back up.

7 Connections to Other AI Models

- Hopfield networks: early associative memory with energy landscapes.
- Boltzmann machines: probabilistic EBMs.
- GANs: adversarial tug-of-war is similar to energy shaping.
- Diffusion models: iterative denoising resembles Langevin sampling. Note that $\nabla_x \log p_\theta(x) = -\nabla_x E_\theta(x)$, so the Langevin update above is a gradient step along the *score*, exactly the quantity that a diffusion noise predictor estimates (Week 5 notes, Topic 2).

8 Takeaway

EBMs are like sculpting a landscape:

- Valleys = real data.
- Mountains = fake data.

Training ensures real examples fall into valleys while fake ones are pushed up.

References

- Y. LeCun, S. Chopra, R. Hadsell, M. Ranzato, F. J. Huang, "A Tutorial on Energy-Based Learning," in *Predicting Structured Data*, MIT Press, 2006.
- Y. Du, I. Mordatch, "Implicit Generation and Modeling with Energy-Based Models," NeurIPS 2019, arXiv:1903.08689.
- Y. Song, D. P. Kingma, "How to Train Your Energy-Based Models," 2021, arXiv:2101.03288.
- D. Foster, *Generative Deep Learning*, 2nd ed., O'Reilly, 2023, Chapter 7 (Energy-Based Models). Companion notebook `notebooks/07_ebm/01_ebm`.